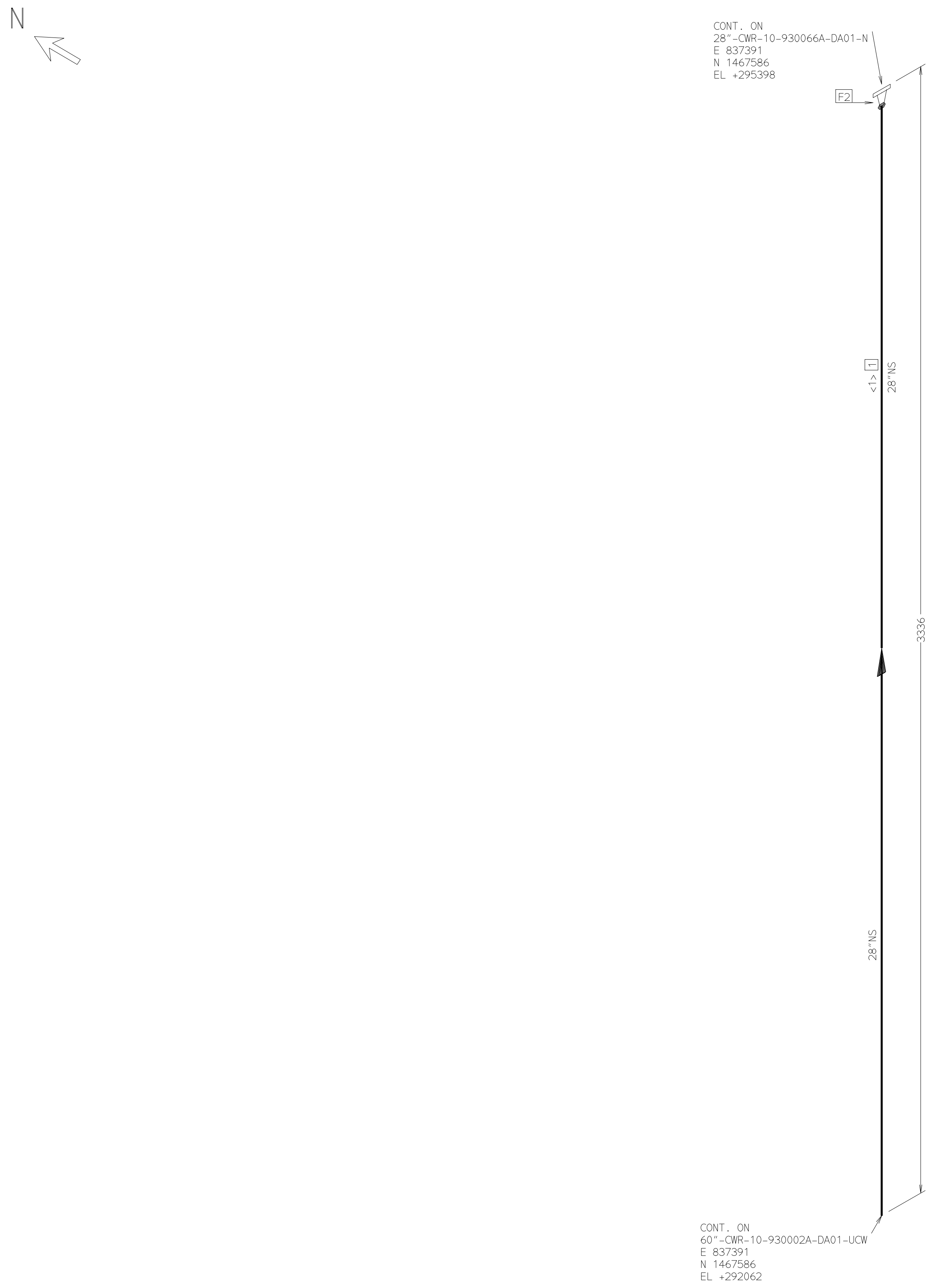


BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E290WL	3.3M
	FLANGES			
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

NOTE

1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION	DESIGN	OPERATING
PRESS. (barg):	7.4	2.3
TEMP.(°C)	65	43
PRESS.TEST(barg): 11.1	TEST FLUID: FRESH WATER	
INSULATION TYPE:	INSULATION THK.: -	
PAINT CODE: -	FIRE PROOFING: NO	
STRESS ANALYSIS:		

TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT
			PWHT: -
	ELEC. TRACING:		PHASE: L

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت پتروشیمی ابن سینای اندیمشک (سهامی خاص)

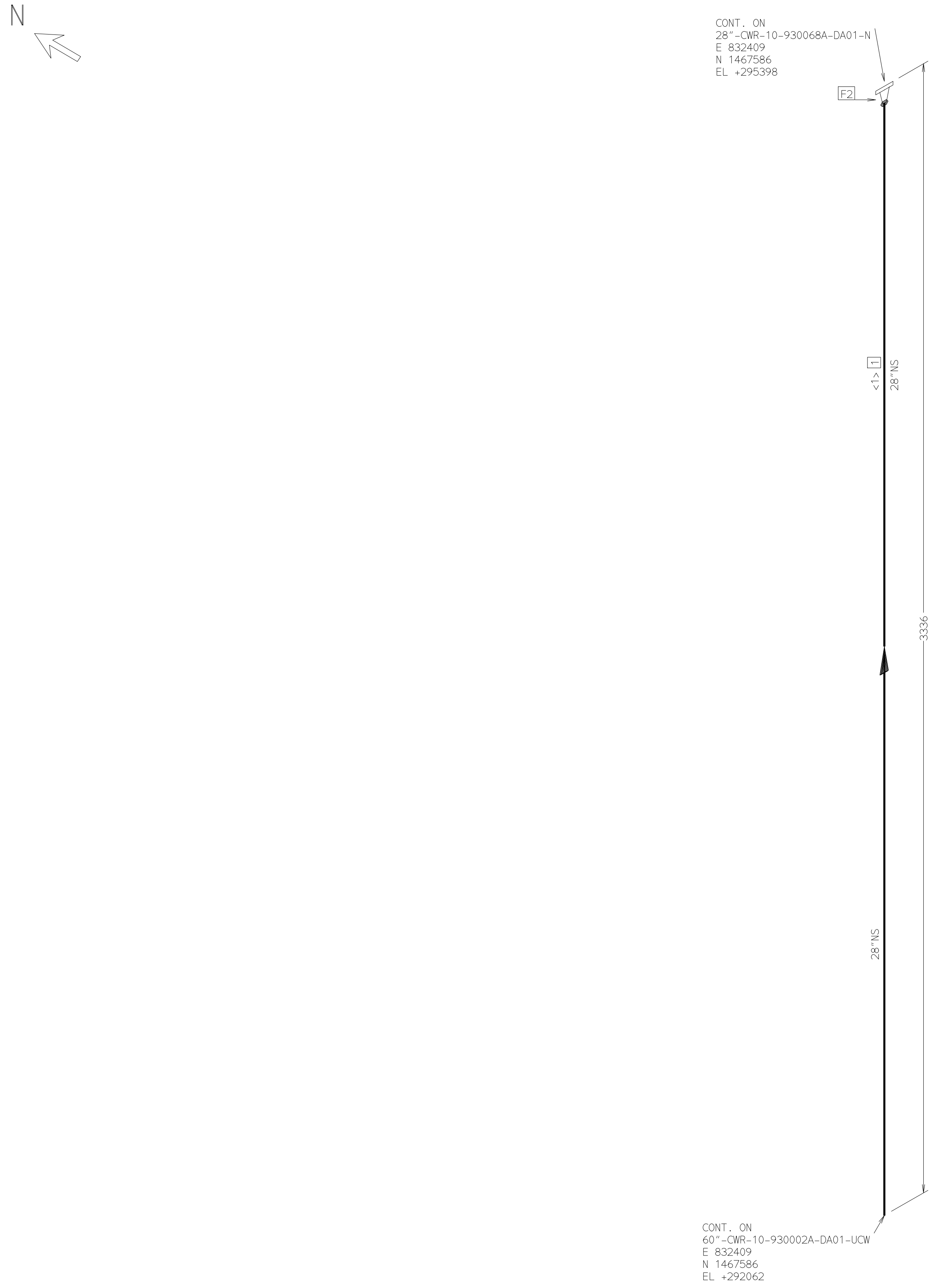
MC:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت مدیریت توسعه صنایع پتروشیمی (سهامی خاص)

CONTRACTOR:
شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
28"-CWR-10-930065A-DA01-UCW	A0	1B	1 OF1

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E290WL	3.3M
	FLANGES			
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

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CONDITION		DESIGN		OPERATING
PRESS. (barg):		7.4		2.3
TEMP.(°C)		65		43
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER		
INSULATION TYPE:			INSULATION THK.: -	
PAINT CODE: -			FIRE PROOFING: NO	
STRESS ANALYSIS:				
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	
			PWHT: -	
	ELEC. TRACING:		PHASE: L	

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت پتروشیمی ابن سینای اندیمشک (سهامی خاص)

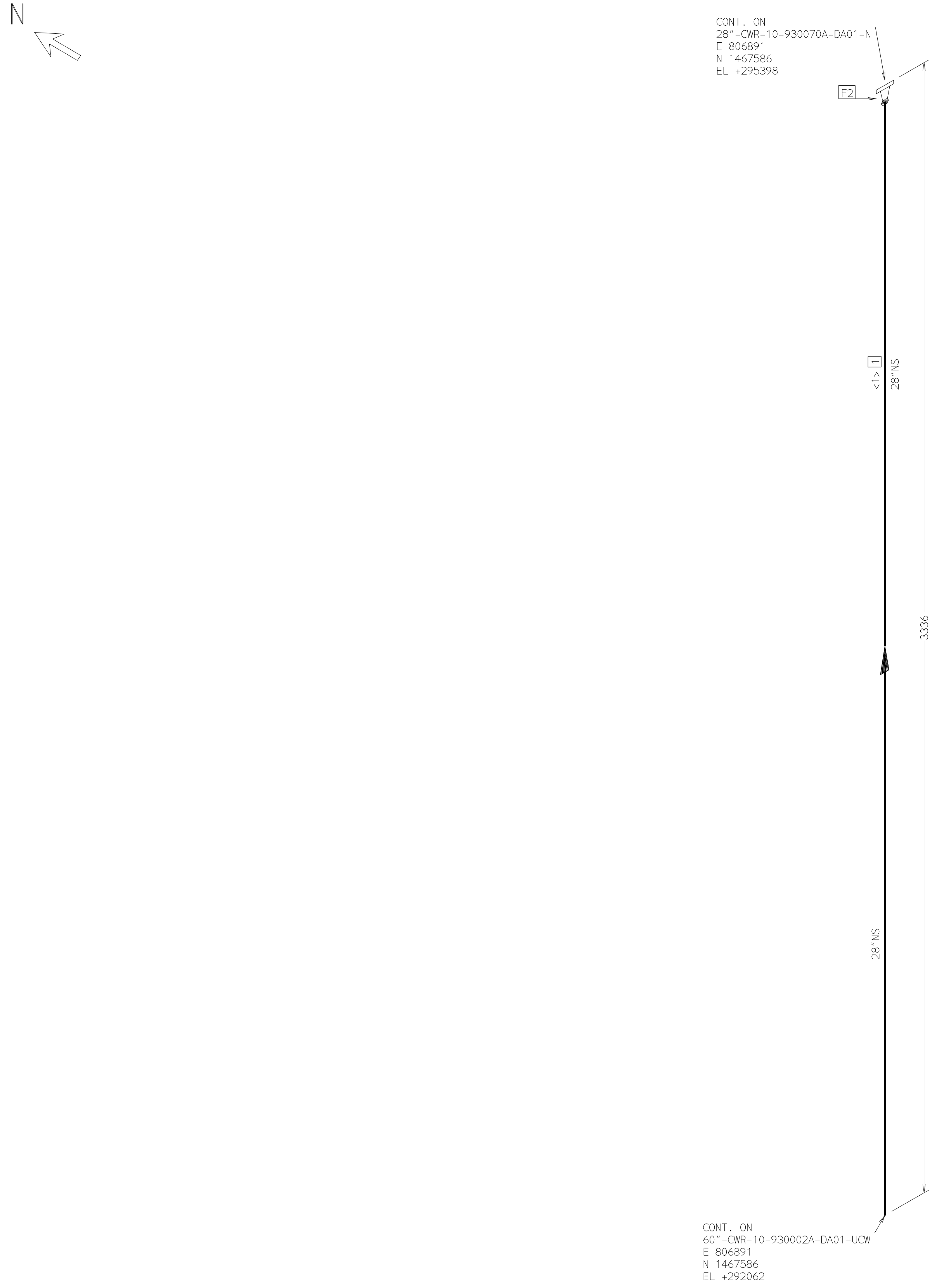
MC:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت مدیریت توسعه صنایع پتروشیمی (سهامی خاص)

CONTRACTOR:
شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
28"-CWR-10-930067A-DA01-UCW	A0	1B	1 OF1

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E290WL	3.3M
	FLANGES			
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

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CONDITION	DESIGN	OPERATING
PRESS. (barg):	7.4	2.3
TEMP.(°C)	65	43
PRESS.TEST(barg): 11.1	TEST FLUID: FRESH WATER	
INSULATION TYPE:	INSULATION THK.: -	
PAINT CODE: -	FIRE PROOFING: NO	
STRESS ANALYSIS:		

TRACING			
STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	
		PWHT: -	
ELEC. TRACING:		PHASE: L	

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

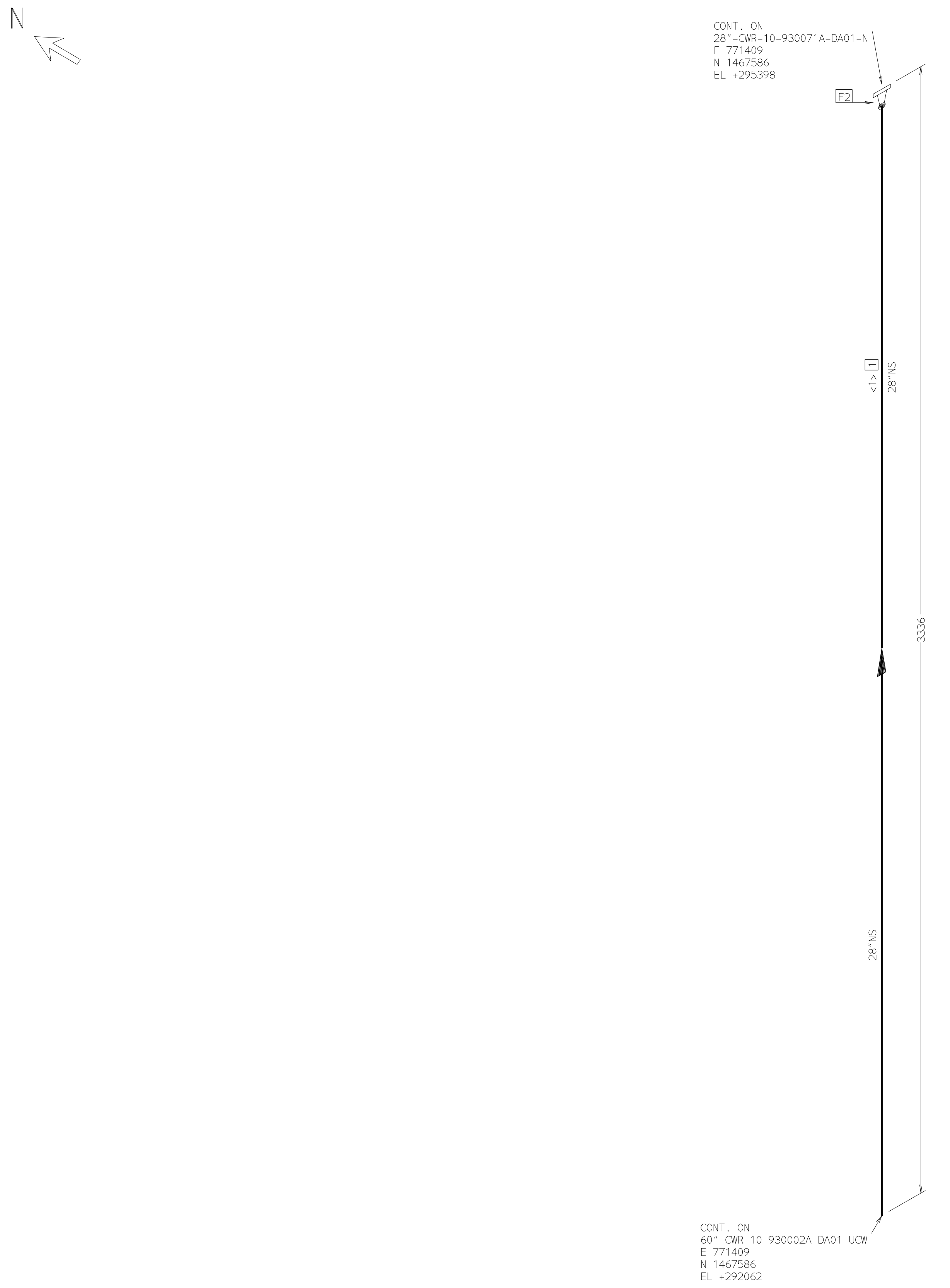
REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:
MC:
CONTRACTOR:
شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
28"-CWR-10-930069A-DA01-UCW	A0	1B	1 OF1

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E290WL	3.3M
	FLANGES			
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

NOTE

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CONDITION		DESIGN	OPERATING
PRESS. (barg):		7.4	2.3
TEMP.(°C)		65	43
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER	
INSULATION TYPE:		INSULATION THK.: -	
PAINT CODE: -		FIRE PROOFING: NO	
STRESS ANALYSIS:			
TRACING	STEAM TRACING QTY. / TRACING SIZE	NDT: 10%RT+10%MT	
		PWHT: -	
	ELEC. TRACING:	PHASE: L	

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت پتروشیمی ابن سینای اندیمشک (سهامی خاص)

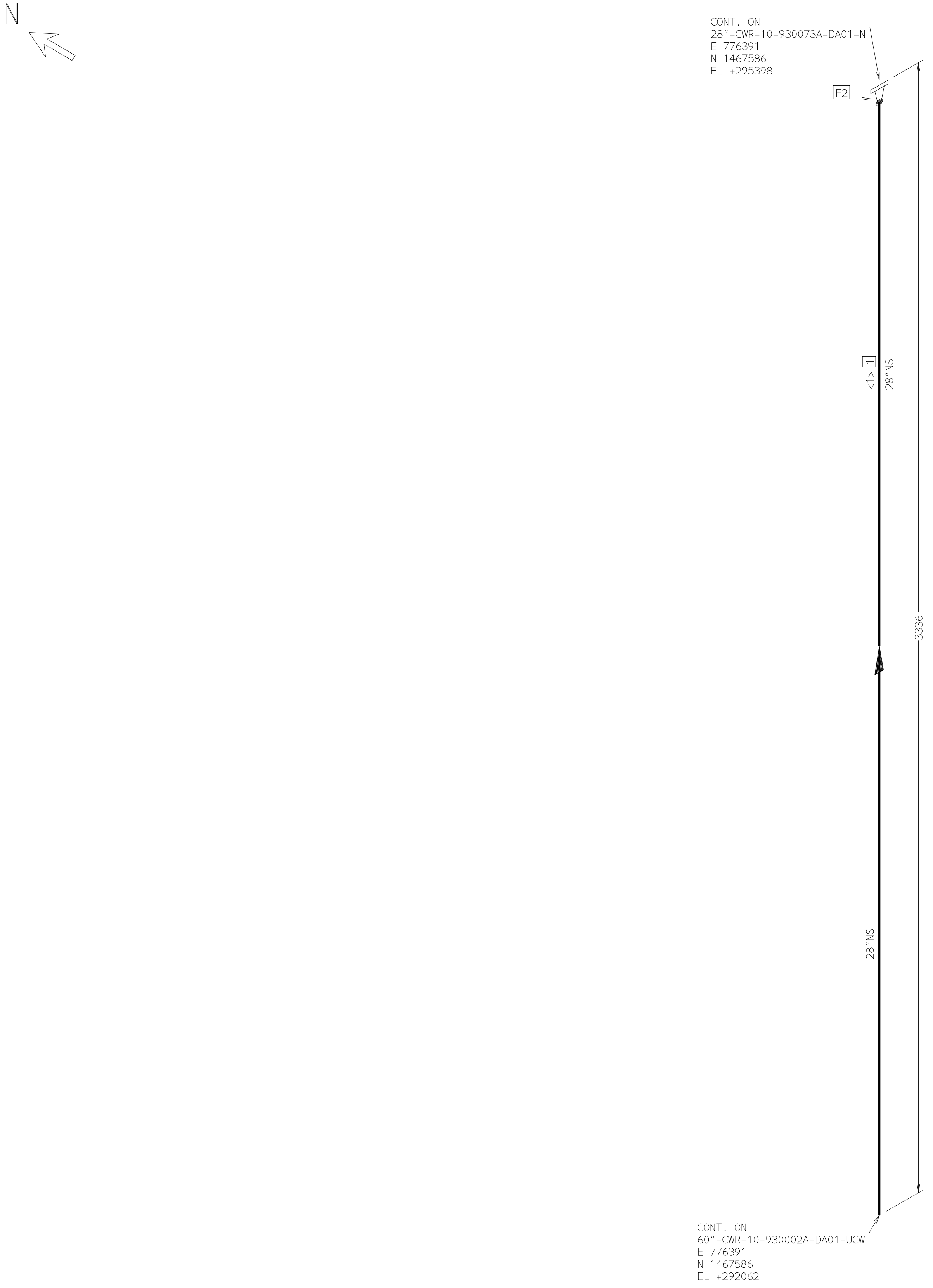
MC:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت مدیریت توسعه صنایع پتروشیمی (سهامی عام)

CONTRACTOR:
شرکت پترو سازه مبین مهندسین مشاور جگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.: 28"-CWR-10-930072A-DA01-UCW
REV.: A0
CLASS: 1B
SHT.NO.: 1 OF1

PROJECT DOCUMENT No.: AISPC-10-DE-930-PI-ISO-0001
PROJECT DOCUMENT TITLE: ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT
REV: A0
AREA No.: 930



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E290WL	3.3M
	FLANGES			
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

NOTE

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CONDITION		DESIGN		OPERATING
PRESS. (barg):		7.4		2.3
TEMP.(°C)		65		43
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER		
INSULATION TYPE:			INSULATION THK.: -	
PAINT CODE: -			FIRE PROOFING: NO	
STRESS ANALYSIS:				
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	
			PWHT: -	
	ELEC. TRACING:		PHASE: L	

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت پتروشیمی ابن سینای اندیمشک (سهامی خاص)

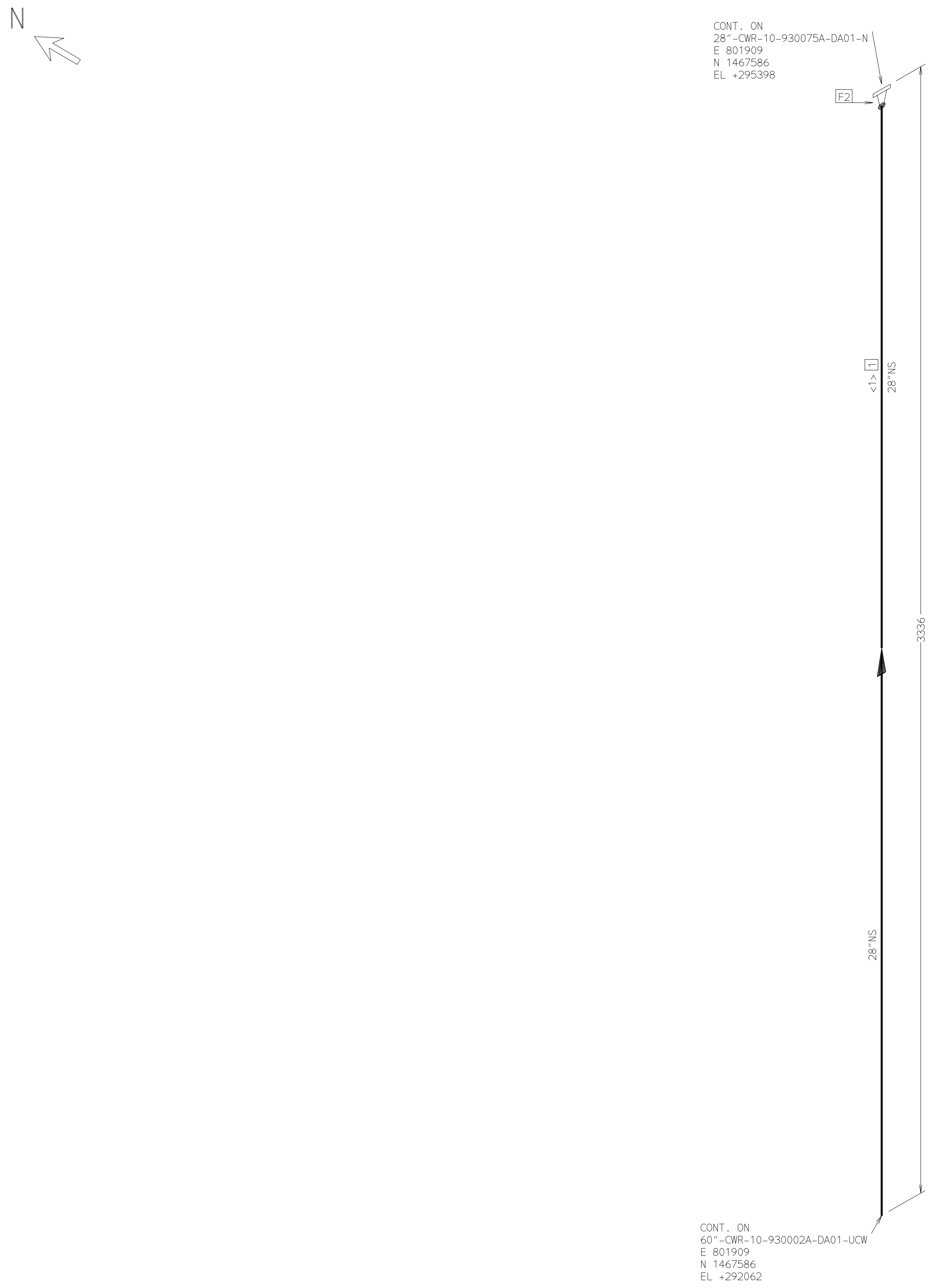
MC:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت مدیریت توسعه صنایع پتروشیمی (سهامی خاص)

CONTRACTOR:
شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
28"-CWR-10-930074A-DA01-UCW	A0	1B	1 OF1

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, SCH 10, ASME B36.10M	28	125A1E290WL	3.3M
	FLANGES			
2	Flange, ASTM A105N, Weld Neck, RF, SCH 10, 150#, ASME B16.47B	28	3625BAQ1CJ	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	3211	28			

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CONDITION	DESIGN	OPERATING
PRESS. (barg):	7.4	2.3
TEMP.(°C)	65	43
PRESS.TEST(barg): 11.1	TEST FLUID: FRESH WATER	
INSULATION TYPE:	INSULATION THK.: -	
PAINT CODE: -	FIRE PROOFING: NO	
STRESS ANALYSIS:		

TRACING			
	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT
			PWHT: -
	ELEC. TRACING:		PHASE: L

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:
Persian Gulf Petrochemical Industries Co
PGPIC

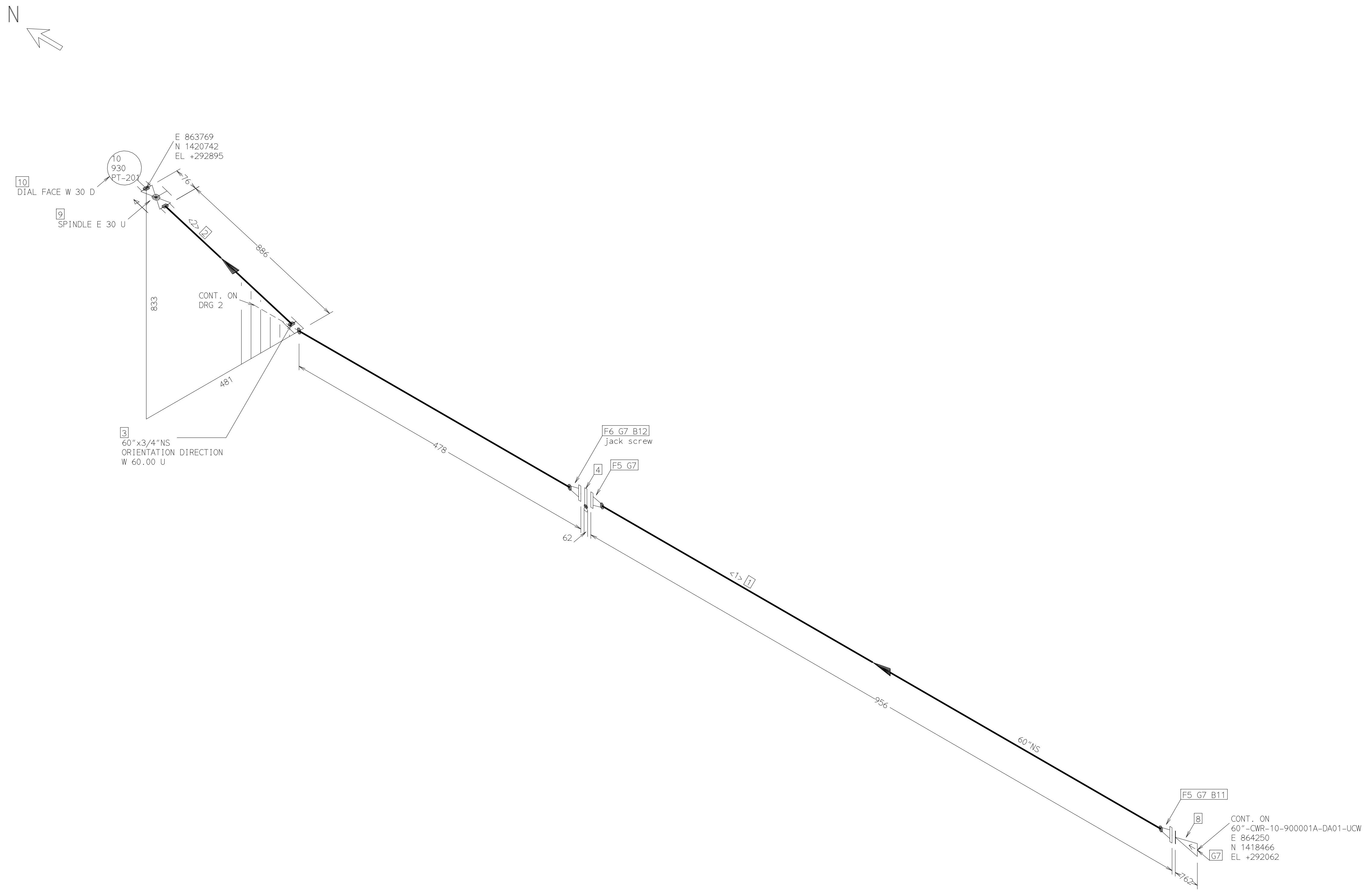
MC:
Persian Gulf Petrochemical Industries Co
PGPIC

CONTRACTOR:
شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
28"-CWR-10-930076A-DA01-UCW	A0	1B	1 OF1

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	0.9M
2	Pipe, ASTM A106 GR.B, Seamless, PE, SCH 80, ASME B36.10M	3/4	16F1K11AL	0.1M
	FITTINGS			
3	Half Coupling, ASTM A105N, SW, 3000#, MSS SP-97	60 x 3/4	246KD1CB	1
4	Blind & Open Spacer, ASTM A516 GR.70, RF, 150#, MFR Standard	60	6441BQ549X	1
	FLANGES			
5	Flange, ASTM A105N, Weld Neck, RF, THK 15.88mm, 150#, ASME B16.47B	60	3641BZQ1CJ	2
6	Flange, ASTM A105N, Weld Neck JACK SCREW, RF, SCH 15.88MM, 150#, ASME B16.47B, AISPC-10-DE-100-PI-SDG-0002	60	6141BZQ1CJ8E	1
	GASKETS			
7	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.47B, ASME B16.20	60	6941B1L7F	4
	VALVES / IN-LINE ITEMS			
8	Butterfly valve, Category A, 150#, Lug type, G.O, API 609/MAN STD/ASME B16.34/ASME B16.47B Body: ASTM A216 Gr.WCB /BU100	60	9C41B1P526UI7R	1
9	Globe Valve, Swivel Plug Type, OS&Y, BB, HO, SW/SCRD, 800#, API 602/ASME B16.34/ASME B16.11 Body: ASTM A105N, Trim No.8/GL115	3/4	46601C1C3DR8R	1
	INSTRUMENTS			
10	INSTRUMENT GAUGE NPT 10 930 PT-201	3/4		1

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	BOLTS			
11	1130 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	1.1/4	547UP4	52
12	330 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	1.1/4	86B2C	52

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)
<1>	600	60
<3>	300	60

PIECE NO	CUT LENGTH	N.S. (INS)
<2>	100	3/4

NOTE

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CONDITION	DESIGN	OPERATING
PRESS. (barg):	7.4	2.3
TEMP.(°C)	65	43

PRESS.TEST(barg): 11.1
TEST FLUID: FRESH WATER
INSULATION TYPE:
INSULATION THK.: -
PAINT CODE: -
FIRE PROOFING: NO
STRESS ANALYSIS:

TRACING			
STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	
		PWHT: -	
ELEC. TRACING: NO		PHASE: L	

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:
Persian Gulf Petrochemical Industries Co
PGPIC

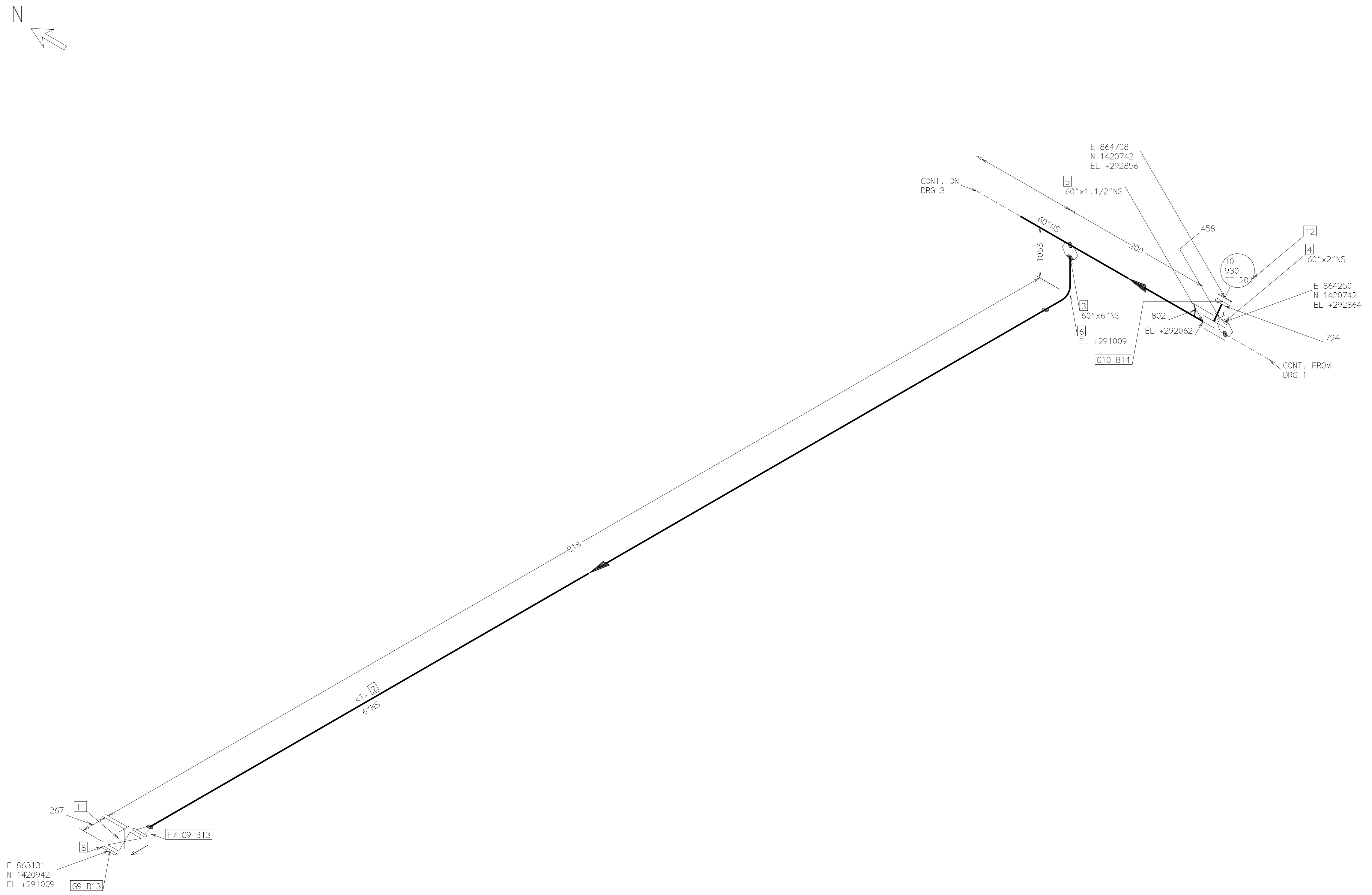
MC:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت مدیریت توسعه صنایع پتروشیمی (سهامی خاص)

CONTRACTOR:
شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
60"-CWR-10-930001A-DA01-UCW	A0	1B	1 OF4

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	0.3M
2	Pipe, ASTM A106 GR.B, Seamless, BE, SCH 40, ASME B36.10M	6	114D1E11AL	0.5M
	FITTINGS			
3	Weldolet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97	60 x 6	254114ZD4M1CU	1
4	Weldolet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97 ASTM A105N	60 x 2	254110ZD4M1CU	1
5	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	1
6	Elbow 90, ASTM A234 WPB, Seamless, LR, BW, SCH 40, ASME B16.9	6	214D4M53AA	1
	FLANGES			
7	Flange, ASTM A105N, Weld Neck, RF, SCH 40, 150#, ASME B16.5	6	3614BDQ1CG	1
8	Flange, Blind , ASTM A105N, RF, 150#, ASME B16.5	6	3814BQ1CG	1
	GASKETS			
9	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.5, ASME B16.20	6	6914B1L2k	2
10	Gasket, SS304/GRAPHITE, IR SS304, OR K.C.S, Spiral Wound, 4.5 mm Thick., RF, 300#, ASME B16.5, ASME B16.20	1.1/2	699D1L2k	1
	VALVES / IN-LINE ITEMS			
11	Gate Valve, Flexible Wedge, BB, OS&Y, HO, FLGD, RF, 150#, API 600/ASME B16.34/ASME B16.5 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	6	4814BQ526PR8R	1
	INSTRUMENTS			
12	INSTRUMENT GAUGE #300 RF 10 930 TT-201	1.1/2		1
PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	BOLTS			
13	110 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BK	16
14	100 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BI	4

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	500	6			

NOTE

1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION		DESIGN	OPERATING
PRESS. (barg):		7.4	2.3
TEMP.(°C)		65	43
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER	
INSULATION TYPE:		INSULATION THK.: -	
PAINT CODE: -		FIRE PROOFING: NO	
STRESS ANALYSIS:			
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT
			PWHT: -
	ELEC. TRACING: NO		PHASE: L

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT: Persian Gulf Petrochemical Industries Co PGPIC

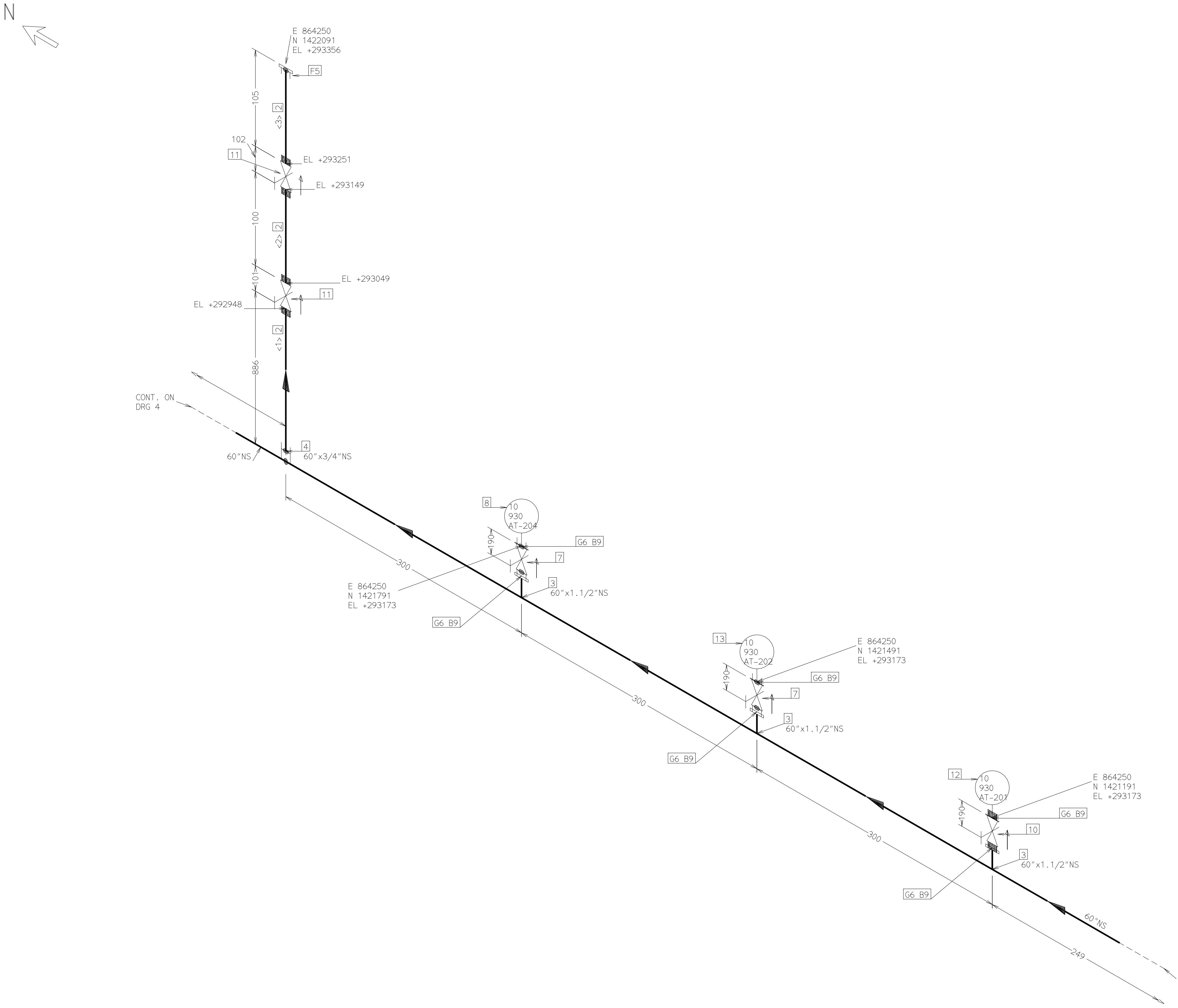
MC: Persian Gulf Petrochemical Industries Co PGPIC

CONTRACTOR: شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT: ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
60"-CWR-10-930001A-DA01-UCW	A0	1B	2 OF4

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930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DRG 4
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4
60"x3/4"NS
8
10
930
AT-204
190
G6 B9
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60"x1.1/2"NS
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G6 B9
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190
G6 B9
10
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60"x1.1/2"NS
300
G6 B9
60"NS
249
CONT. FROM
DRG 2

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	1.2M
2	Pipe, ASTM A106 GR.B, Seamless, PE, SCH 80, ASME B36.10M	3/4	16F1K11AL	0.3M
	FITTINGS			
3	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	3
4	Half Coupling, ASTM A105N, SW, 3000#, MSS SP-97	60 x 3/4	246KD1CB	1
	FLANGES			
5	Flange, ASTM A105N, Socket welded, RF, SCH80, 150#, ASME B16.5	3/4	356BFQ1CG	1
	GASKETS			
6	Gasket, SS304/GRAPHITE, IR SS304, OR K.C.S, Spiral Wound, 4.5 mm Thick., RF, 300#, ASME B16.5, ASME B16.20	1.1/2	699D1L2k	6
	VALVES / IN-LINE ITEMS			
7	Gate Valve, Solid Wedge, BB, OS&Y, HO, FLGD, RF, B16.5, 300#, API 600 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	1.1/2	489BQ1C8PR8R	2
	INSTRUMENTS			
8	INSTRUMENT GAUGE #300 RF 10 930 AT-204	1.1/2		1
PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	BOLTS			
9	100 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BI	24
	VALVES / IN-LINE ITEMS			
10	Gate Valve, Solid Wedge, BB, OS&Y, HO, FLGD, RF, B16.5, 300#, API 600 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	1.1/2	489BQ1C8PR8R	1
11	Gate Valve, Solid Wedge, BB, OS&Y, HO, SW, 800#, API 602 Body: ASTM A105N, Trim No.8/GA122	3/4	4860D1C3DR8R	2
	INSTRUMENTS			
12	INSTRUMENT GAUGE #300 RF 10 930 AT-201	1.1/2		1
13	INSTRUMENT GAUGE #300 RF 10 930 AT-202	1.1/2		1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	448	60	<2>	299	60
<3>	299	60	<4>	100	3/4
<5>	100	3/4	<6>	100	3/4

NOTE

1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION	DESIGN	OPERATING
PRESS. (barg):	7.4	2.3
TEMP.(°C)	65	43
PRESS.TEST(barg): 11.1	TEST FLUID: FRESH WATER	
INSULATION TYPE:	INSULATION THK.: -	
PAINT CODE: -	FIRE PROOFING: NO	
STRESS ANALYSIS:		

TRACING			
STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	
		PWHT: -	
ELEC. TRACING: NO		PHASE: L	

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:

MC:

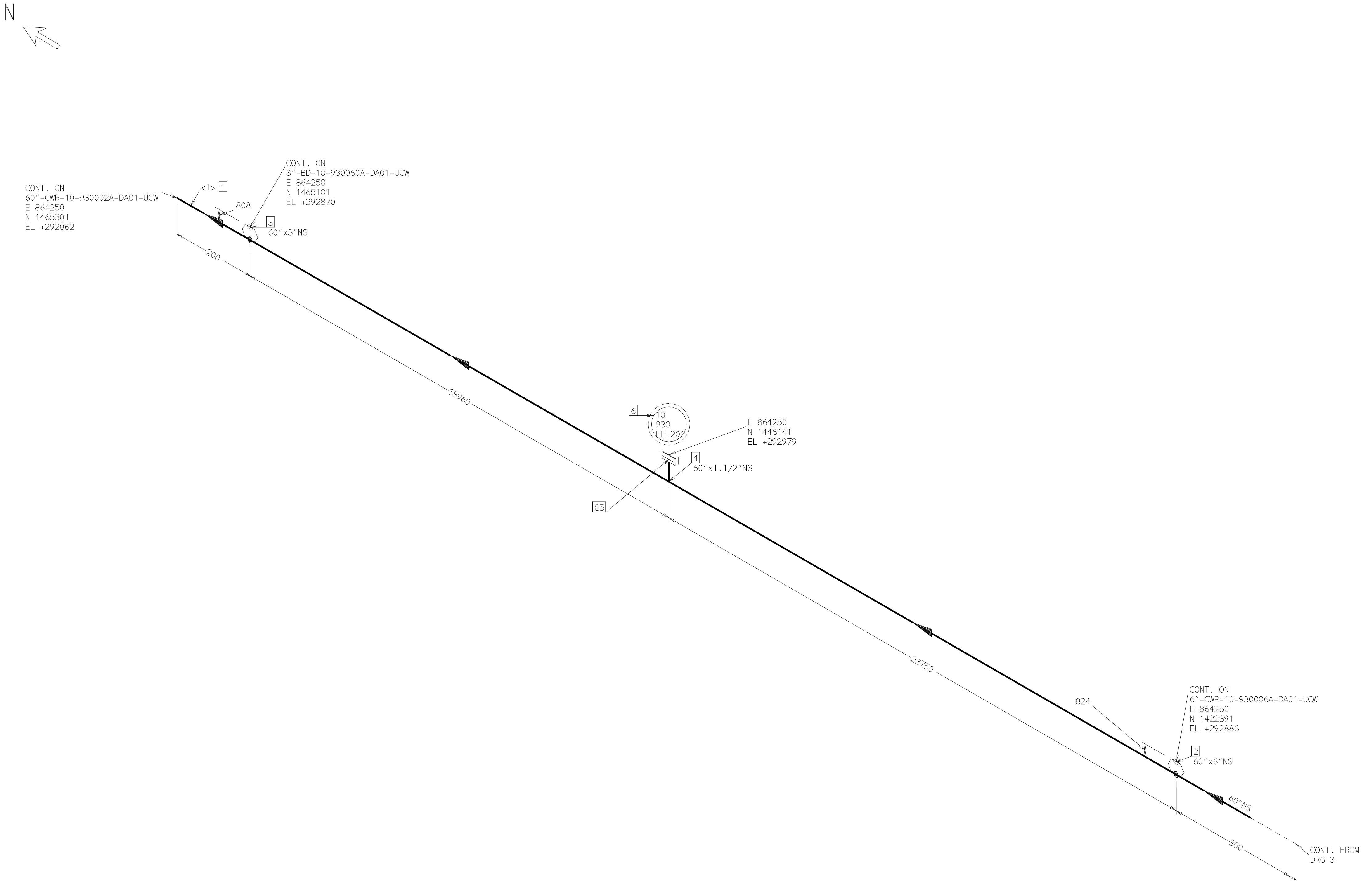
CONTRACTOR:

شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
60"-CWR-10-930001A-DA01-UCW	A0	1B	3 OF4

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930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60"x1.1/2"NS
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60"NS
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CONT. FROM
DRG 3

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	43.3M
	FITTINGS			
2	Weldolet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97	60 x 6	254114ZD4M1CU	1
3	Weldolet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97 ASTM A105N	60 x 3	254112ZD4M1CU	1
4	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	1
	GASKETS			
5	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.5, ASME B16.20	1.1/2	699B1L2k	1
	INSTRUMENTS			
6	INSTRUMENT GAUGE #150 RF 10 930 FE-201	1.1/2		1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	24349	60	<2>	19160	60

NOTE

1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
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5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION	DESIGN	OPERATING
PRESS. (barg):	7.4	2.3
TEMP.(°C)	65	43
PRESS.TEST(barg): 11.1	TEST FLUID: FRESH WATER	
INSULATION TYPE:	INSULATION THK.: -	
PAINT CODE: -	FIRE PROOFING: NO	
STRESS ANALYSIS:		

TRACING			
STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	
		PWHT: -	
ELEC. TRACING: NO		PHASE: L	

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت پتروشیمی ابن سینای اندیمشک (سهامی خاص)

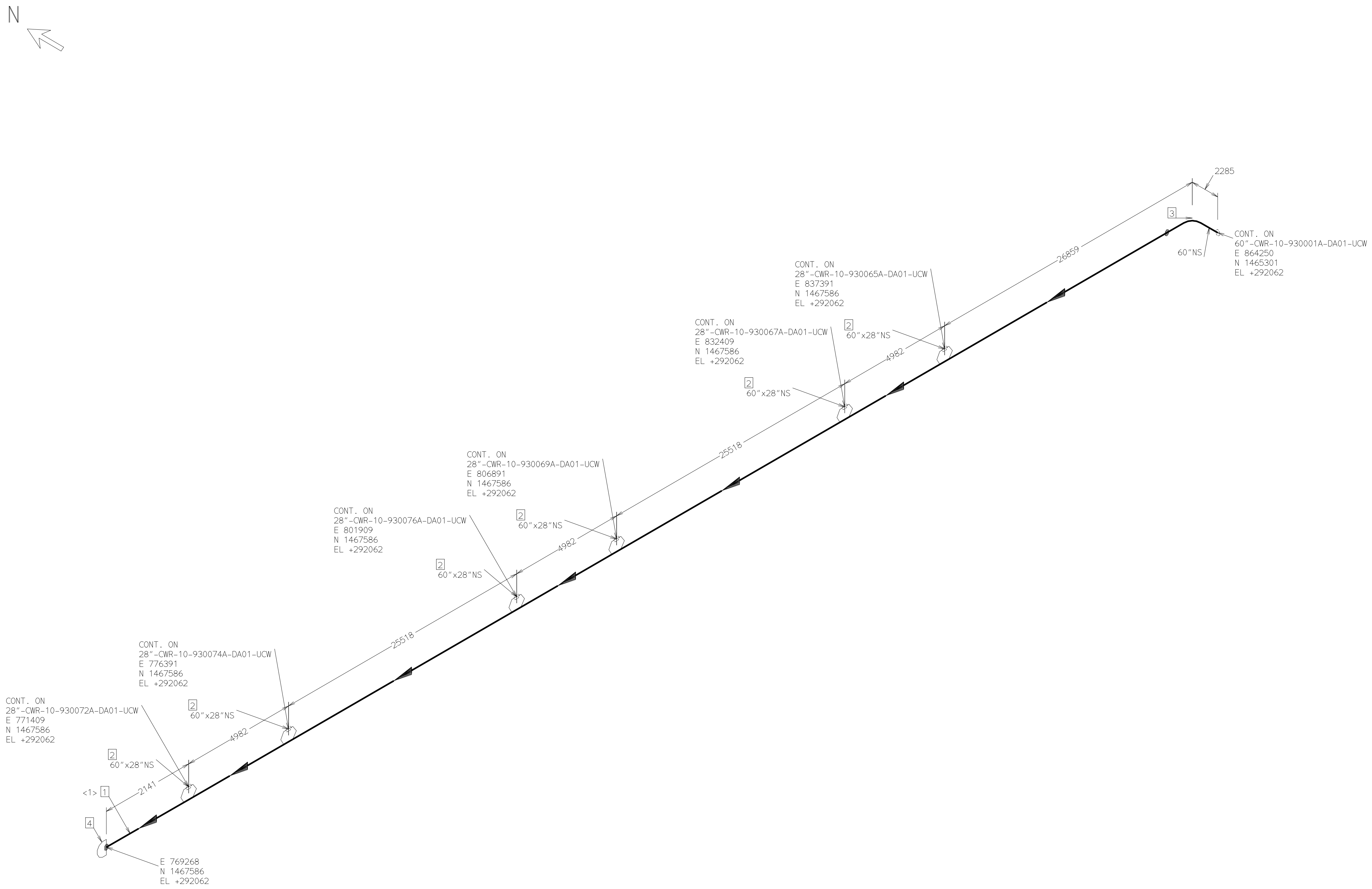
MC:
شرکت صنایع پتروشیمی خلیج فارس
Persian Gulf Petrochemical Industries Co
PGPIC
شرکت مدیریت توسعه صنایع پتروشیمی (سهامی خاص)

CONTRACTOR:
شرکت پترو سازه مبین مهندسین مشاور جگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
60"-CWR-10-930001A-DA01-UCW	A0	1B	4 OF4

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930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EL +292062

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	92.7M
	FITTINGS			
2	PIPE TO PIPE	60 x 28	-	6
3	Elbow 90, ASTM A234 WPB, Welded, LR, BW, THK 15.88mm, ASME B16.9/MAN STD	60	241Z4M530W2Q	1
4	Cap, ASTM A234 WPB, Welded, BW, SCH 15.88MM, ASME B16.9	60	1741Z4M53A2Q	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	92697	60			

NOTE

1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
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7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION	DESIGN	OPERATING
PRESS. (barg):	7.4	2.3
TEMP.(°C)	65	43
PRESS.TEST(barg): 11.1	TEST FLUID: FRESH WATER	
INSULATION TYPE:	INSULATION THK.: -	
PAINT CODE: -	FIRE PROOFING: NO	
STRESS ANALYSIS:		

TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT
			PWHT: -
	ELEC. TRACING: NO		PHASE: L

A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO
REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:

Persian Gulf Petrochemical Industries Co
PGPIC

MC:

Persian Gulf Petrochemical Industries Co
PGPIC

CONTRACTOR:

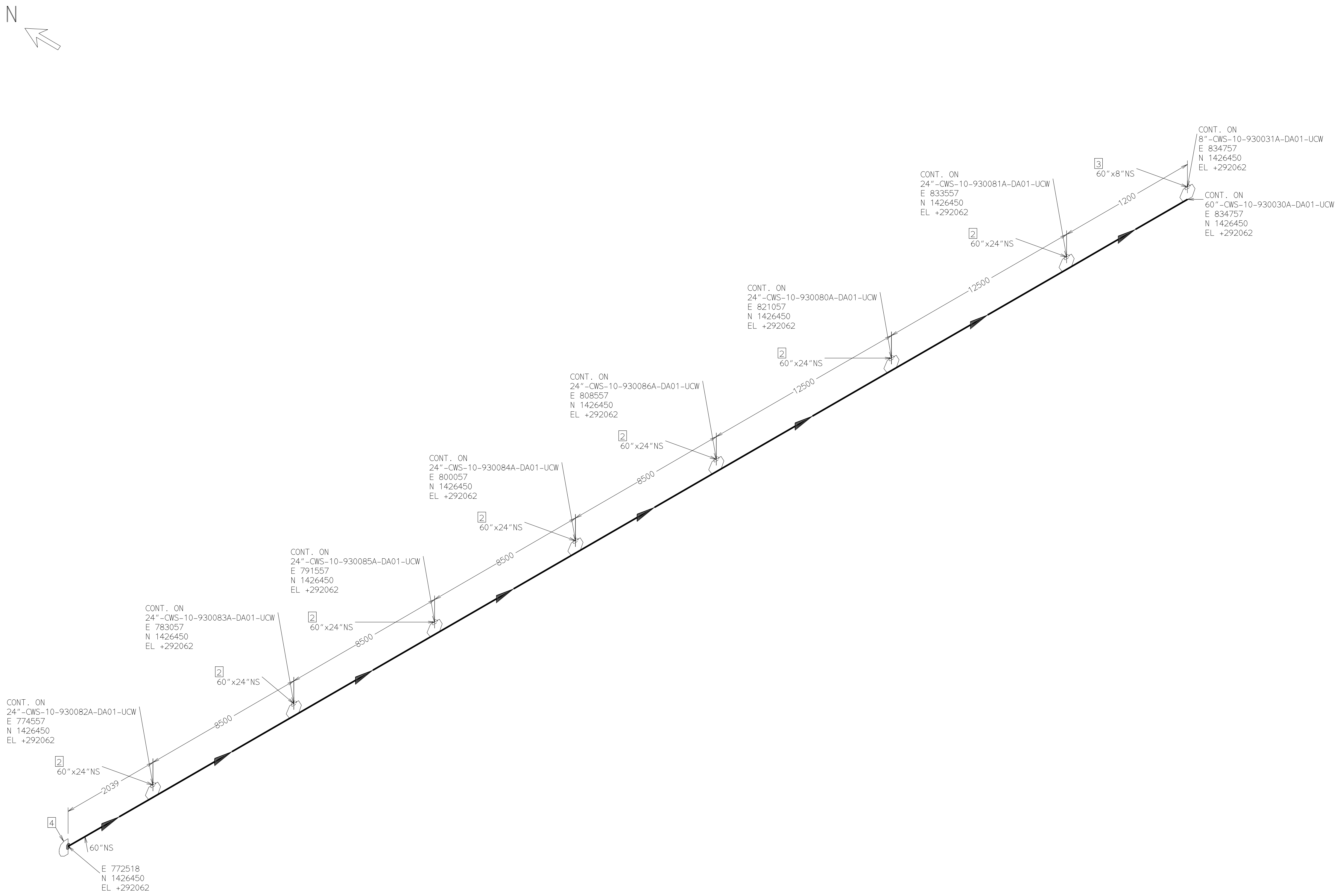
شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:

ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
60"-CWR-10-930002A-DA01-UCW	A0	1B	1 OF1

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930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8500
CONT. ON
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8500
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N 1426450
EL +292062
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8500
CONT. ON
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N 1426450
EL +292062
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60"NS
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N 1426450
EL +292062

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	62.3M
	FITTINGS			
2	PIPE TO PIPE	60 x 24		7
3	PIPE TO PIPE	60 x 8		1
4	Cap, ASTM A234 WPB, Welded, BW, SCH 15.88MM, ASME B16.9	60	1741Z4M53A2Q	1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	62239	60			

NOTE

1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
3-FOR $1\frac{1}{2}$" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
6-ISOMETRIC DRAWING FOR $1\frac{1}{2}$" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION		DESIGN		OPERATING
PRESS. (barg):		7.4		5.9
TEMP.(°C)		65		33
PRESS.TEST(barg): 11.1		TEST FLUID: FRESH WATER		
INSULATION TYPE:			INSULATION THK.: -	
PAINT CODE: -			FIRE PROOFING: NO	
STRESS ANALYSIS:				
TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT	
			PWHT: N	
	ELEC. TRACING: NO		PHASE: L	

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:

MC:

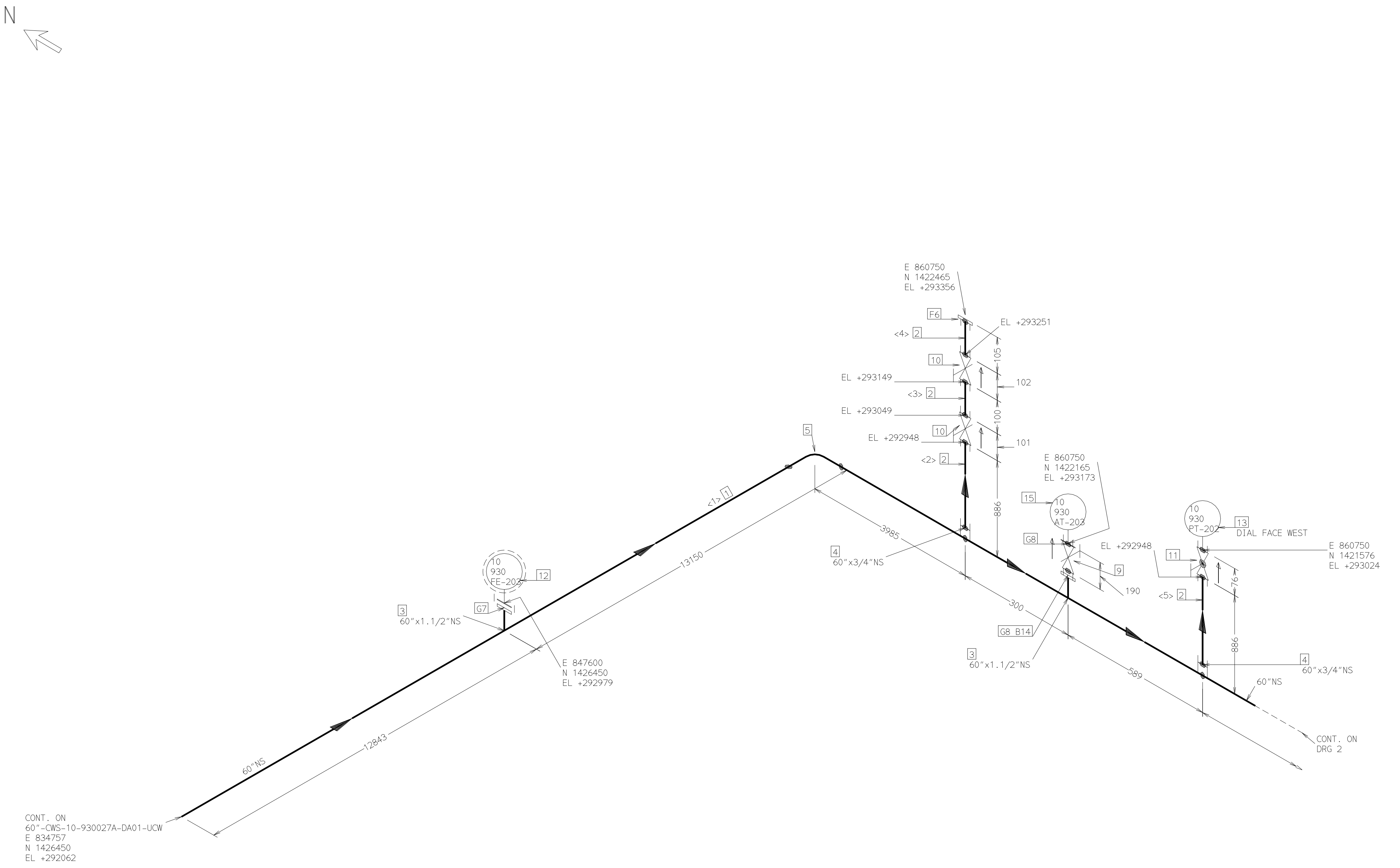
CONTRACTOR:

شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
60"-CWS-10-930027A-DA01-UCW	A0	1B	1 OF1

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930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60"-CWS-10-930027A-DA01-UCW
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N 1426450
EL +292062

BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	26.3M
2	Pipe, ASTM A106 GR.B, Seamless, PE, SCH 80, ASME B36.10M	3/4	16F1K11AL	0.4M
	FITTINGS			
3	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	2
4	Half Coupling, ASTM A105N, SW, 3000#, MSS SP-97	60 x 3/4	246KD1CB	2
5	Elbow 90, ASTM A234 WPB, Welded, LR, BW, THK 15.88mm, ASME B16.9/MAN STD	60	241Z4M530W2Q	1
	FLANGES			
6	Flange, ASTM A105N, Socket welded, RF, SCH80, 150#, ASME B16.5	3/4	356BFQ1CG	1
	GASKETS			
7	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.5, ASME B16.20	1.1/2	699B1L2k	1
8	Gasket, SS304/GRAPHITE, IR SS304, OR K.C.S, Spiral Wound, 4.5 mm Thick., RF, 300#, ASME B16.5, ASME B16.20	1.1/2	699D1L2k	2
	VALVES / IN-LINE ITEMS			
9	Gate Valve, Solid Wedge, BB, OS&Y, HO, FLGD, RF, B16.5, 300#, API 600 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	1.1/2	489BQ1C8PR8R	1
10	Gate Valve, Solid Wedge, BB, OS&Y, HO, SW, 800#, API 602 Body: ASTM A105N, Trim No.8/GA122	3/4	4860D1C3DR8R	2
11	Globe Valve, Swivel Plug Type, OS&Y, BB, HO, SW/SCRD, 800#, API 602/ASME B16.34/ASME B16.11 Body: ASTM A105N, Trim No.8/GL115	3/4	46601C1C3DR8R	1
	INSTRUMENTS			
12	INSTRUMENT GAUGE #150 RF 10 930 FE-203	1.1/2		1
13	INSTRUMENT GAUGE NPT 10 930 PT-202	3/4		1
PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	BOLTS			
14	100 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BI	4
	INSTRUMENTS			
15	INSTRUMENT GAUGE #150 RF 10 930 AT-203	1.1/2		1

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	12842	60	<2>	10865	60
<3>	100	3/4	<4>	100	3/4
<5>	100	3/4	<6>	1999	60
<7>	100	3/4			

NOTE

1-EXTRA PIPE SHALL BE CONSIDERED FOR FIELD WELD ADJUSTMENT ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
2-ALL CUTTING PIPE LENGTHS ARE FOR INFORMATION ONLY AND SHALL BE ON 3 DIRECTORY BY CONSTRUCTION CONTRACTOR.
3-FOR 1½" AND SMALLER PIPING'S SUPPORTS TO BE SELECTED BY CONSTRUCTION CONTRACTOR EXCEPT AS NOTED.
4-ACTUAL SIZE OF ALL INSTRUMENT ITEM AND PIPING SPECIAL ITEMS SHALL BE CHECKED BEFORE ANY SPOOL FABRICATION ACTIVITIES.
5-ROOT GAPS OF BUTT-WELDING FITTINGS ARE NOT INCLUDED IN DIMENSIONS.
6-ISOMETRIC DRAWING FOR 1½" AND SMALLER PIPING'S ARE JUST FOR INFORMATION PIPING ROUTING DIMENSIONS AND ELEVATION CHECKED BY CONSTRUCTION CONTRACTOR BEFORE INSTALLATION.
7-FIRST REVISION OF ISOMETRIC DOCUMENTS MUST BE APPROVED BEFORE CONSTRUCTION.

CONDITION	DESIGN	OPERATING
PRESS. (barg):	7.4	5.9
TEMP.(°C)	65	33
PRESS.TEST(barg): 11.1	TEST FLUID: FRESH WATER	
INSULATION TYPE:	INSULATION THK.: -	
PAINT CODE: -	FIRE PROOFING: NO	
STRESS ANALYSIS:		

TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT
			PWHT: N
	ELEC. TRACING: NO		PHASE: L

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:

Persian Gulf Petrochemical Industries Co
PGPIC

MC:

Persian Gulf Petrochemical Industries Co
PGPIC

CONTRACTOR:

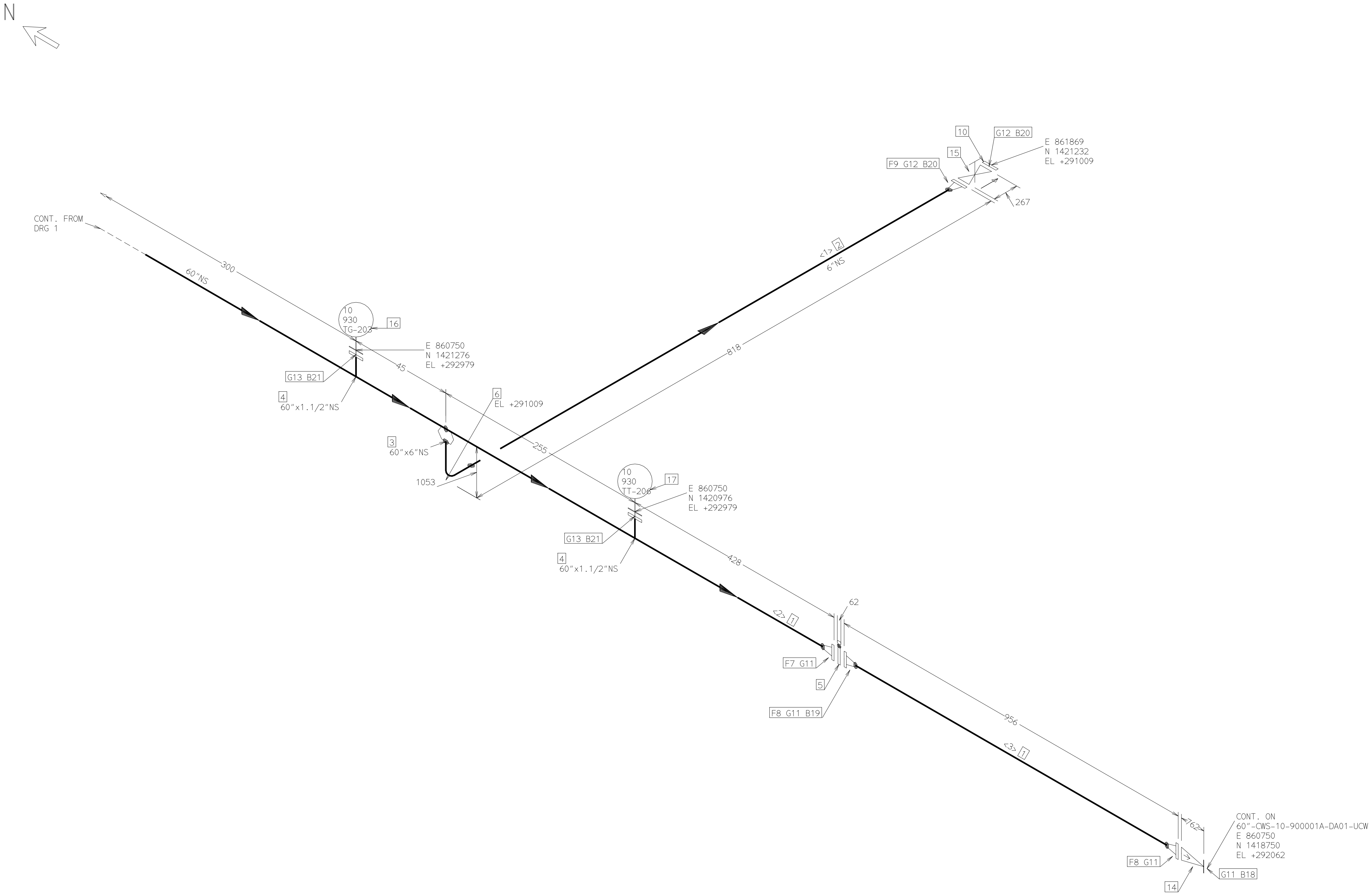
شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:

ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
60"-CWS-10-930030A-DA01-UCW	A0	1B	1 OF2

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT	A0	930



BILL OF MATERIAL

PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	PIPE			
1	Pipe, API 5L GR.B 100% RT, LSAW, BE, THK 15.88mm, MAN STD	60	141Z1E290WU	1.5M
2	Pipe, ASTM A106 GR.B, Seamless, BE, SCH 40, ASME B36.10M	6	114D1E11AL	0.5M
	FITTINGS			
3	Weldolet, ASTM A105N, BW, THK 15.88mmxSCH 40, MSS SP-97	60 x 6	254114ZD4M1CU	1
4	Flange, Long Weld Neck , RF, BW, SCH 80, 300#, ASTM A105N, ASME B16.5	60 x 1.1/2	41419DFQ1CG	2
5	Blind & Open Spacer, ASTM A516 GR.70, RF, 150#, MFR Standard	60	6441BQ549X	1
6	Elbow 90, ASTM A234 WPB, Seamless, LR, BW, SCH 40, ASME B16.9	6	214D4M53AA	1
	FLANGES			
7	Flange, ASTM A105N, Weld Neck JACK SCREW, RF, SCH 15.88MM, 150#, ASME B16.47B, AISPC-10-DE-100-PI-SDG-0002	60	6141BZQ1CJ8E	1
8	Flange, ASTM A105N, Weld Neck, RF, THK 15.88mm, 150#, ASME B16.47B	60	3641BZQ1CJ	2
9	Flange, ASTM A105N, Weld Neck, RF, SCH 40, 150#, ASME B16.5	6	3614BDQ1CG	1
10	Flange, Blind , ASTM A105N, RF, 150#, ASME B16.5	6	3814BQ1CG	1
	GASKETS			
11	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.47B, ASME B16.20	60	6941B1L7F	4
12	Gasket, Spiral Wound, SS304/GRAPHITE ,IR SS304,OR K.C.S 4.5 mm Thick., RF, 150#, ASME B16.5, ASME B16.20	6	6914B1L2k	2
13	Gasket, SS304/GRAPHITE, IR SS304, OR K.C.S, Spiral Wound, 4.5 mm Thick., RF, 300#, ASME B16.5, ASME B16.20	1.1/2	699D1L2k	2
	VALVES / IN-LINE ITEMS			
14	Butterfly valve, Category A, 150#, Lug type, G.O, API 609/MAN STD/ASME B16.34/ASME B16.47B Body: ASTM A216 Gr.WCB /BU100	60	9C41B1P526UI7R	1
15	Gate Valve, Flexible Wedge, BB, OS&Y, HO, FLGD, RF, 150#, API 600/ASME B16.34/ASME B16.5 Body: ASTM A216 Gr.WCB, Trim No.8/GA101	6	4814BQ526PR8R	1
	INSTRUMENTS			
16	INSTRUMENT GAUGE #300 RF 10 930 TG-203	1.1/2		1
17	INSTRUMENT GAUGE #300 RF 10 930 TT-206	1.1/2		1
PT NO	COMPONENT DESCRIPTION	N.S. (INS)	ITEM CODE	QTY
	BOLTS			
18	1130 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	1.1/4	547UP4	52
19	330 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	1.1/4	86B2C	52
20	110 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BK	16
21	100 Stud Bolt, A193 B7 A194 2H DACROMET, FULL THD, two heavy HEX nuts, ASME B18.2.1/ASME B18.2.2/ASME B1.1	3/4	66BI	8

CUT PIPE LENGTH

PIECE NO	CUT LENGTH	N.S. (INS)	PIECE NO	CUT LENGTH	N.S. (INS)
<1>	888	60	<2>	500	6
<3>	299	60	<4>	250	60
<5>	600	60			

NOTE

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CONDITION	DESIGN	OPERATING
PRESS. (barg):	7.4	5.9
TEMP.(°C)	65	33
PRESS.TEST(barg): 11.1	TEST FLUID: FRESH WATER	
INSULATION TYPE:	INSULATION THK.: -	
PAINT CODE: -	FIRE PROOFING: NO	
STRESS ANALYSIS:		

TRACING	STEAM TRACING QTY.	TRACING SIZE	NDT: 10%RT+10%MT
			PWHT: N
	ELEC. TRACING: NO		PHASE: L

REV.	DATE	DESCRIPTION	ISSUED BY	CHECKED BY	APPROVED BY
A0	06/JAN/2026	ISSUED FOR REVIEW	M.IMANZADEH	M.MOHAMMADI	F.IZADJOO

REF. P&ID DWG. No.: AISPC-10-DE-930-PR-PID-0001-(1 of 11)

CLIENT:
MC:
CONTRACTOR:
شرکت پترو سازه مبین مهندسین مشاور چگالش

PROJECT:
ANDIMESHK IBN SINA PETROCHEMICAL COMPANY
ETHYLENE OXIDE PLANT

LINE No.:	REV:	CLASS:	SHT.NO.
60"-CWS-10-930030A-DA01-UCW	A0	1B	2 OF2

PROJECT DOCUMENT No.:	PROJECT DOCUMENT TITLE:	REV:	AREA No.:
AISPC-10-DE-930-PI-ISO-0001	**ISOMETRIC DRAWINGS FOR UNDERGROUND COOLING WATER UNIT**	A0	930